Year 11 Mathematics

Number 9

6.3 Compound Percentages

Compound Percentages

STARTER · 5 MIN

Date: _____ **Title:** Compound Percentages

LEARNING OBJECTIVES

- Calculate simple interest
- Calculate compound interest and depreciation
- Solve problems involving repeated percentage change

STARTER *Answer in your book. No calculators.*

1. Work out 1.1×1.1
2. Write the multiplier for a 4% increase
3. Write the multiplier for a 10% decrease
4. Work out 2^3

Starter — Answers

STARTER

Tick or correct in green pen

1. **1.21**

2. **1.04**

3. **0.9**

4. $2 \times 2 \times 2 = \mathbf{8}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can calculate simple interest	☐
SILVER	I can calculate compound interest and depreciation	☐
GOLD	I can solve problems involving repeated percentage change	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Compound Percentages

INTRODUCTION · 3 MIN

Simple interest is the same every year. Compound interest earns interest on the interest.

Simple interest

The same interest every year.

£100 at 5%: £5 each year

Compound interest

Interest is added each year.

£100 × 1.05^3

Depreciation

The value falls each year.

£800 × 0.85^3

Formula

Original × multiplier to the power n.

n = number of years

Examples 1 and 2 — Simple Interest

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint interest = rate × amount × years

Example 1

£500 is invested at 4% simple interest for 3 years. Find the interest.

Example 2

£2000 is invested at 3% simple interest for 5 years. Find the total amount.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

4% of 500 = £20

20 × 3

£60

Tip: find one year, then multiply.

Example 2

3% of 2000 = £60

60 × 5 = £300

2000 + 300 **= £2300**

Tip: total = original + interest.

Task 1

PRACTICE · 9 MIN

BRONZE *Simple interest*

Hint interest = rate × amount × years

1. £400 at 5% for 2 years: find the interest
2. £1000 at 3% for 4 years: find the interest
3. £250 at 2% for 6 years: find the interest
4. £800 at 4% for 3 years: find the total
5. £1500 at 6% for 2 years: find the total

★ £600 at 2.5% for 4 years: find the total

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $20 \times 2 = \mathbf{£40}$
2. $30 \times 4 = \mathbf{£120}$
3. $5 \times 6 = \mathbf{£30}$
4. $800 + 96 = \mathbf{£896}$
5. $1500 + 180 = \mathbf{£1680}$

★ $600 + 60 = \mathbf{£660}$

Examples 3 and 4 — Compound Interest and Depreciation

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint value = original × multiplier^{n}

Example 3

£500 is invested at 4% compound interest for 3 years. Find the total amount to the nearest penny.

Example 4

A car worth £12 000 depreciates by 15% a year. Find its value after 4 years to the nearest penny.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Multiplier = 1.04

500×1.04^3

£562.43

Tip: the power is the number of years.

Example 4

Multiplier = 0.85

$12\,000 \times 0.85^4$

£6264.08

Tip: depreciate = decrease.

Task 2

PRACTICE · 9 MIN

SILVER

Give answers to the nearest penny

Hint value = original × multiplier^{n}

1. £2000 at 3% compound interest for 2 years
2. £800 at 5% compound interest for 3 years
3. £5000 at 2.5% compound interest for 4 years
4. A £900 laptop loses 20% of its value each year. Find its value after 3 years.
5. A population of 4000 grows by 6% a year. Find it after 5 years.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $2000 \times 1.03^2 = \mathbf{£2121.80}$
2. $800 \times 1.05^3 = \mathbf{£926.10}$
3. $5000 \times 1.025^4 = \mathbf{£5519.06}$
4. $900 \times 0.8^3 = \mathbf{£460.80}$
5. $4000 \times 1.06^5 = \mathbf{5353}$

Examples 5 and 6 — Compound Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint value = original × multiplier^{n}

Example 5

(3 marks)

£3000 is invested at 4% compound interest. After how many years will it first be more than £3500?

Example 6

(3 marks)

£1000 for 3 years: 5% simple interest or 4.5% compound interest. Which gives more?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$3000 \times 1.04^3 = £3374.59$

$3000 \times 1.04^4 = £3509.58$

4 years

Tip: try values of n.

Example 6

Simple: 1000 + 150 = £1150

Compound: $1000 \times 1.045^3 = £1141.17$

Simple, by £8.83

Tip: work out both totals.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working. Give money to the nearest penny.

1. £4000 is invested at 3.5% compound interest for 3 years. Find the interest earned. (3 marks)

2. A house worth £180 000 increases in value by 4% a year. Find its value after 5 years. (2 marks)

3. A boat worth £25 000 loses 12% of its value each year. Find its value after 3 years. (2 marks)

4. £2000 is invested at 5% compound interest. After how many years is it more than £2500? (3 marks)

5. A population of 50 000 falls by 2% a year. After how many years is it below 45 000? (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $4000 \times 1.035^3 = £4434.87$, $4434.87 - 4000 =$ **£434.87**

2. $180\,000 \times 1.04^5 =$ **£218 997.53**

3. $25\,000 \times 0.88^3 =$ **£17 036.80**

4. $2000 \times 1.05^4 = £2431.01$, $2000 \times 1.05^5 = £2552.56$, **5 years**

5. $50\,000 \times 0.98^5 = 45\,196$, $50\,000 \times 0.98^6 = 44\,292$, **6 years**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE £200 at 5% simple interest for 3 years: find the interest

SILVER Write the multiplier for 7% compound growth

GOLD £1000 at 10% compound interest for 2 years: find the total

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $10 \times 3 = \mathbf{£30}$

SILVER **1.07**

GOLD $1000 \times 1.1^2 = \mathbf{£1210}$

Exam Practice — Simple Interest

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint interest = rate × amount × years

Question 1 (2 marks)

£600 is invested at 3% simple interest for 4 years.

Work out the interest.

..... £

Question 2 (3 marks)

£2500 is invested at 2.5% simple interest for 6 years.

Work out the total amount at the end.

..... £

Exam Practice — Compound Interest and Depreciation

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint value = original × multiplier^{n}

Question 3

(3 marks)

£3000 is invested at 3.5% compound interest for 4 years.

Work out the total amount.

Give your answer to the nearest penny.

..... £

Question 4

(3 marks)

A van is worth £22 000.

It depreciates by 18% each year.

Work out its value after 3 years.

Give your answer to the nearest pound.

..... £

Exam Practice — Compound Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint value = original × multiplier^{n}

Question 5

(3 marks)

£5000 is invested at 4% compound interest.

After how many whole years will it first be worth more than £6000?

..... years

Question 6

(3 marks)

A population of 80 000 falls by 3% each year.

After how many whole years will it first be below 70 000?

..... years

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

3% of 600 = £18 a year — M1

18 × 4 = £72 — A1

Tip: find one year, then multiply.

Question 2

2.5% of 2500 = £62.50 — M1

62.50 × 6 = £375 — M1

2500 + 375 = £2875 — A1

Tip: total = original + interest.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

multiplier = 1.035 — M1

3000×1.035^4 — M1

= £3442.57 — A1

Tip: the power is the number of years.

Question 4

multiplier = 0.82 — M1

$22\,000 \times 0.82^3$ — M1

= £12 130 — A1

Tip: depreciate means decrease.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$5000 \times 1.04^4 = £5849.29$ **M1**

$5000 \times 1.04^5 = £6083.26$ **M1**

5 years **A1**

Tip: try values of n.

Question 6

$80\,000 \times 0.97^4 = 70\,823\ldots$ **M1**

$80\,000 \times 0.97^5 = 68\,699\ldots$ **M1**

5 years **A1**

Tip: show the year before and the year after.